Cross-Validation Under Linear Truth

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Plan

In this project, I simulate data where the true relationship is exactly linear and then compare two candidate models with repeated cross-validation:

- **Linear (correct): `y ~ x`**
- **Polynomial degree 3 (more flexible): `y ~ poly(x, 3)`**

The goal is to show that even with very large sample sizes, cross-validation does not guarantee selecting the true linear model in every finite run.

Methods and replication

```r
library(tidyverse)

set.seed(2026)

K <- 10L
cv_repeats <- 3L

sample_sizes <- c(10000L, 30000L)
mc_runs_by_n <- c(30L, 6L)
beta0 <- 1
beta1 <- 2
sigma <- 4
poly_degree <- 3L
```

The data-generating process is:

- $x \sim \mathcal{N}(0, 1)$
- $y = \beta_0 + \beta_1 x + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$

For each sample size and each Monte Carlo run:

1. Simulate one dataset under the linear truth.
2. Repeat 10-fold CV multiple times using fresh random folds.
3. Compute mean CV MSE for linear and polynomial models.
4. Select the model with lower repeated-CV mean MSE.

```
run_one_dataset <- function(n, K, cv_repeats, beta0, beta1, sigma, poly_degree) {
  x <- rnorm(n)
  eps <- rnorm(n, sd = sigma)
  y <- beta0 + beta1 * x + eps
  dat <- data.frame(y = y, x = x)

  rep_mean_cv_linear <- numeric(cv_repeats)
  rep_mean_cv_poly <- numeric(cv_repeats)

  for (r in seq_len(cv_repeats)) {
    fold_id <- sample(rep(seq_len(K), length.out = n))
    mse_linear <- numeric(K)
    mse_poly <- numeric(K)

    for (k in seq_len(K)) {
      train <- fold_id != k
      val <- fold_id == k

      fit_lin <- lm(y ~ x, data = dat, subset = train)
      fit_poly <- lm(
        y ~ poly(x, degree = poly_degree, raw = TRUE),
        data = dat,
        subset = train
      )

      pred_lin <- predict(fit_lin, newdata = dat[val, , drop = FALSE])
      pred_poly <- predict(fit_poly, newdata = dat[val, , drop = FALSE])

      y_val <- dat$y[val]
      mse_linear[k] <- mean((y_val - pred_lin)^2)
      mse_poly[k] <- mean((y_val - pred_poly)^2)
    }
```

```r
    rep_mean_cv_linear[r] <- mean(mse_linear)
    rep_mean_cv_poly[r] <- mean(mse_poly)
  }

  mean_cv_linear <- mean(rep_mean_cv_linear)
  mean_cv_poly <- mean(rep_mean_cv_poly)
  cv_gap_poly_minus_linear <- mean_cv_poly - mean_cv_linear

  selected_model <- if (mean_cv_linear <= mean_cv_poly) "linear" else "poly3"

  tibble(
    mean_cv_linear = mean_cv_linear,
    mean_cv_poly3 = mean_cv_poly,
    cv_gap_poly_minus_linear = cv_gap_poly_minus_linear,
    selected_model = selected_model
  )
}
```

```r
all_results <- vector("list", length(sample_sizes))

for (i in seq_along(sample_sizes)) {
  n_now <- sample_sizes[i]
  mc_runs_now <- mc_runs_by_n[i]
  run_results <- vector("list", mc_runs_now)

  for (m in seq_len(mc_runs_now)) {
    run_results[[m]] <- run_one_dataset(
      n = n_now,
      K = K,
      cv_repeats = cv_repeats,
      beta0 = beta0,
      beta1 = beta1,
      sigma = sigma,
      poly_degree = poly_degree
    ) |>
      mutate(n = n_now, mc_run = m)
  }

  all_results[[i]] <- bind_rows(run_results)
}

results <- bind_rows(all_results) |>
```

```r
relocate(n, mc_run)
```

Results

Table 1: Model selection frequency over Monte Carlo runs for each sample size.

n	selected_model	count	selection_rate
10000	linear	22	0.7333
10000	poly3	8	0.2667
30000	linear	5	0.8333
30000	poly3	1	0.1667

Table 2: Average repeated-CV MSE and gap summary by sample size (poly3 minus linear).

n	mean_cv_linear	mean_cv_poly3	mean_gap_poly_minus_linear	sd_gap_poly_minus_linear
10000	15.96091	15.96380	0.00289	0.00423
30000	16.02094	16.02189	0.00095	0.00103

Table 3: How often CV selects the polynomial model even though linear is the truth.

n	poly3_selection_rate	poly3_count
10000	0.2667	8
30000	0.1667	1

Across the large-sample settings in this simulation, the polynomial model is selected in a nonzero share of Monte Carlo runs. This confirms that repeated cross-validation can still choose the wrong model in finite samples, even when the true data-generating relationship is linear and sample size is large.

Conclusion

This simulation isolates model-selection behavior under a known linear truth. Even with repeated 10-fold cross-validation and very large sample sizes, model choice is still random at the margin when candidate models have very similar predictive risk. As a result, cross-validation is highly useful but not guaranteed to recover the true model in every realization.